REVIEW

Let p, q, and r be the propositions

p : Grizzly bears have been seen in the area.
q : Hiking is safe on the trail.
r : Berries are ripe along the trail.

Write these propositions using p, q, and r and logical connectives (including negations).

a) Berries are ripe along the trail, but grizzly bears have not been seen in the area.

b) Grizzly bears have not been seen in the area and hiking on the trail is safe, but berries are ripe along the trail.

c) If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area.

Suppose $L(x,y)$ is the statement

"**x** buys Christmas gifts for **y**",

where **x** and **y** are members of a family.

Translate the statement into logical expression

"*Each family member buys Christmas gifts for any other member but not for himself or herself*"

(i) $\exists x(\neg L(x,x))$

(ii) $\forall x(\neg L(x,x))$

(iii) $\forall x(\forall y L(x,y))$

(iv) $\forall x(\forall y((y \neq x) \leftrightarrow L(x,y)))$

Determine whether these biconditionals are true or false.

a) $2 + 2 = 4$ if and only if $1 + 1 = 2$.
b) $1 + 1 = 2$ if and only if $2 + 3 = 4$.
c) $1 + 1 = 3$ if and only if monkeys can fly.
d) $0 > 1$ if and only if $2 > 1$.

What is the value of x after each of these statements is encountered in a computer program, if $x = 1$ before the statement is reached?

a) **if** $x + 2 = 3$ **then** $x := x + 1$

b) **if** $(x + 1 = 3)$ OR $(2x + 2 = 3)$ **then** $x := x + 1$

c) **if** $(2x + 3 = 5)$ AND $(3x + 4 = 7)$ **then** $x := x + 1$

d) **if** $(x + 1 = 2)$ XOR $(x + 2 = 3)$ **then** $x := x + 1$

e) **if** $x < 2$ **then** $x := x + 1$

Find the bitwise *OR*, bitwise *AND*, and bitwise *XOR* of each of these pairs of bit strings.

a) 101 1110, 010 0001
b) 1111 0000, 1010 1010
c) 00 0111 0001, 10 0100 1000
d) 11 1111 1111, 00 0000 0000

15. For each of these arguments determine whether the argument is correct or incorrect and explain why.

a) All students in this class understand logic. Xavier is a student in this class. Therefore, Xavier understands logic.

b) Every computer science major takes discrete mathematics. Natasha is taking discrete mathematics. Therefore, Natasha is a computer science major.

c) All parrots like fruit. My pet bird is not a parrot. Therefore, my pet bird does not like fruit.

d) Everyone who eats granola every day is healthy. Linda is not healthy. Therefore, Linda does not eat granola every day.

Which of the following option is true?

A. 4 – 2 = 0 whenever 5 + 1 = 6

B. Monkeys will fly, if the Sun is a planet

C. 12>15 and 6 is a positive integer

D. 4 < - 2 and 8 is a negative integer

Let p: This is a great club, q: You should not come back here.

Then “This is a great club and you should come back here.” is best represented by

A. $\neg p \wedge \neg q$

B. $p \wedge q$

C. $p \wedge \neg q$

D. $p \vee \neg q$

Which of the following bits is the negation of the bits “01 0110 0011”?

A. 10 1001 1100

B. 11 1001 0011

C. 00 1001 1100

D. 01 1001 0011

Let A and B be sets such that |B|=|A|+1. Assume that the set AxB has 90 elements. Find |A|?

A. 7 B. 8 C. 9 D. 10

What is the negation of the statement A –> (B v C)?

A. A –> B –> C

B. –A ^ B v C

C. A ^ B v C

D. A ^ –B ^ –C

What are the contrapositive of the conditional statement “I come to class whenever there is going to be a test.

A. If I come to class, then there will be a test.

B. If there is not going to be a test, then I don’t come to class

C. If I do not come to class, then there will not be a test.

10. The $g^{-1}(\{0\})$ for the function $g(x)= [x]$ is _____

a) $\{x \mid 0 \leq x < 1\}$

b) $\{x \mid 0 < x \leq 1\}$

c) $\{x \mid 0 < x < 1\}$

d) $\{x \mid 0 \leq x \leq 1\}$

“Villa is out for a trip or it is not snowing” and “It is snowing or Keane is playing chess” imply that

A. Villa is out for trip.

B. Villa is playing chess

C. Villa is out for a trip and Keane is playing chess

D. Villa is out for a trip or Keane is playing chess.

The statement, “Every comedian is funny”

where C(x) is “x is a comedian” and

F (x) is “x is funny” and the domain consists of all people.

(A) $\exists x\big(C(x) \wedge F(x)\big)$

(B) $\forall x\big(C(x) \wedge F(x)\big)$

(C) $\exists x\big(C(x) \rightarrow F(x)\big)$

(D) $\forall x\big(C(x) \rightarrow F(x)\big)$

(E) $\exists x\big(F(x) \rightarrow C(x)\big)$

Consider the following statements

p: David should exercise.

q: David is not a decent table tennis player.

r: David wants to play good table tennis.

The symbolic form of “David is not a decent table tennis player and if he wants to play good table tennis then he should exercise.” is

A. r –> p ^ q

B. p –> q –> r

C. q ^ (r –>p)

D. q <–> p ^ r